

Emma Rafkin

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Education

Georgetown University

M.S. Computational Linguistics. GPA 4.0

Master's Research Paper: Investigating the Cross-Lingual Transferability of NLP Tasks

Washington, DC

2023-2026

Dartmouth College

B.A. Major: Computer Science, Minor: Linguistics. GPA 3.74, Cum Laude

Hanover, NH

2017-2021

Experience

Johns Hopkins University Applied Physics Laboratory

Laurel, MD

Natural Language Processing Researcher

Aug 2021 – Present

- Created tools for U.S. Government that inject domain knowledge into language models and extract critical information from text
- Awarded 8 grants to pursue research in AI trustworthiness, concept erasure, summarization, entity disambiguation, LLM domain adaptation, AI red teaming, and multimodal anomaly detection
- Led research teams of 3-6 people, presented results to internal and external stakeholders

DALI Lab

Hanover, NH

Software Developer

Jan 2019 – Jun 2021

- Produced full-stack web and mobile applications for external stakeholders every 10 weeks
- Created and led classes on web design and development
- Developer lead, Women in Science Program lead and core executive board member for 2020-2021

Alarm.com

Tysons, VA

Software Engineer Intern

Jun 2020 – Aug 2020

- Created client-facing automated reports for property managers about all devices in their units

Cambly Inc.

San Francisco, CA

Software Engineer Intern

Jan 2020 – Mar 2020

- Enhanced Android application to include English as a Second Language courses for adults.

Dartmouth College Computer Science Department

Hanover, NH

Teaching Assistant

Sept 2018 – Dec 2018

- Graded coursework and tests for OOP course, taught weekly review class to 10-15 students

Featured Research Projects

Investigating the Cross-Lingual Transferability of NLP tasks: Explores the cross-lingual transfer of different task representations in LLMs as language distance increases between source and target languages. (Paper in progress)

Task Arithmetic with Support Languages for Low-Resource ASR: Incorporates information from genetically related high-resource languages to improve performance on low-resource automatic speech recognition models. [\[System Paper\]](#), to be published in an ACL workshop.

CaTE: Guidance on data curation methods for trustworthy AI development. [\[Full Report\]](#) [\[Paper on domain knowledge elicitation\]](#)

Skills

ML/NLP: PyTorch, Tensorflow, Huggingface, training/fine-tuning, activation probing/steering, distributed training, Slurm

Software: Python, JavaScript, TypeScript, React, Java, Kotlin, SQL, ElasticSearch, MongoDB, Git